Granular Synthesis: Conception and Continuity

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Abstract

There are a multitude of computer programs available to perform various tasks using granular synthesis. These programs were not spontaneous creations; they grew from an intellectual tree of knowledge that has been slowly developing for over 200 years. Inside each program resides the ideas of Jean-Baptiste Joseph Fourier, Albert Einstein, Erwin Schrödinger, Dennis Gabor, Iannis Xenakis, Curtis Roads, Barry Truax and others. These composers and scientists are the ghosts in the granular synthesis machine.

1 Introduction

In 1952 Karlheinz Stockhausen wrote to Herbert Eimert stating: "the possibility of realising a *sound-atom* is completely beyond me" [17]. Stockhausen was in a small studio basement in Paris working on his *Study on One Sound*, but his results were unsatisfactory. His dilemma came from the fact that his equipment was not adequate, although he did pinpoint an idea that still interests many musicians today.

The conceptualisation of the sound atom and microsound through granular synthesis has been a long time in development. This paper will identify the contributors to these ideas, and the role they played in identifying key aspects of granular synthesis. It will outline this intellectual tree of knowledge.

2 Knowledge: The Sound Atom

The origins of Granular Synthesis came from the combination of two very fundamental scientific ideas. One idea came from the beginning of the 19th century, the other from the beginning of the 20th century (See Figure 1). Whilst each idea has its own ghosts, I will start from these places as they are milestones from their own specific areas.

At the beginning of the 19th century, Jean-Baptiste Joseph Fourier formulated a theory which has become a key component in many parts of digital sound synthesis. His theory stated that any waveform could be deconstructed into a combination of simple sine waves. The practices of the Fourier Transform (FT) have since been refined and improved, yet the initial idea is still implicit [26].

One hundred years later, Albert Einstein established beyond reasonable doubt, the reality of atoms. Many previous people had thought about the existence of finite particles that combined to make up the world. Some of the ancient Greek philosophers unsuccessfully proposed the idea. Isaac Beeckman toyed with it. Isaac Newton proposed a more definite concept of a particle based view of the world which was, for a short time, accepted but then later rejected. However, it was Einstein's insight and development of the work from Max Planck that established the theory for what was to become known as quantum physics.

As part of this work, Einstein specifically proved that sound and light are made of discrete particles. Niels Bohr developed the Einstein atom into a physical model. Louis de Broglie and Wolfgang Pauli developed the idea on the behaviour of atoms. Werner Heisenberg, and then Paul Dirac worked out the probability matrices from which atomic behaviour can be predicted, and then Erwin Schrödinger took all of these ideas and related them to more easily understood physical ideas. His equation that explained quantum phenomena was so neat and self explanatory it set the way for a whole new breed of quantum physicists [5][11][8][32].

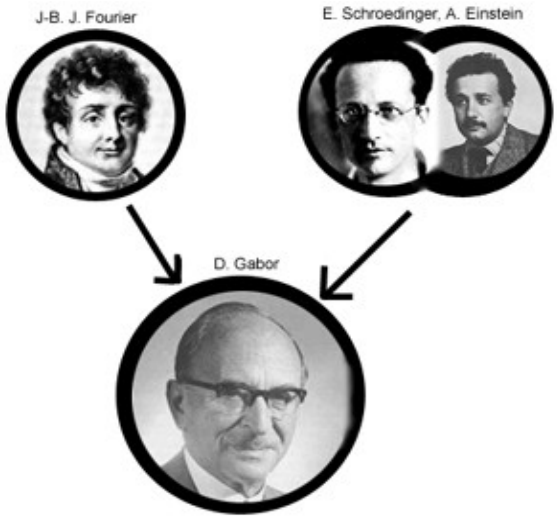


Figure 1: The primary knowledge base of granular synthesis.

Twenty years later at a conference amongst the Institute of Electrical Engineers, Dennis Gabor proposed an idea for reducing the bandwidth required for communication. Gabor was concerned that FT took no account of time. FT worked only in the frequency domain, with the expectation that the

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sound remained unchanged for an infinite amount of time. Using quantum physics as a guide he decided to quantise the sound. He segmented the sound into discrete portions along the time-domain, and then used FT to analyse the frequency domain, which he put into a grid called the Gabor Matrix. Viewing the matrix he could discard all unused output, send only the time segmented information across the bandwidth and then reconstitute it using an inverse FT within the correct time-domain segment. This idea has been used somewhat in signal analysis, data reduction and compression methods, but it is his definition and research into the division of sound, and how to divide it, that has mostly concerned musical composers [10].

The quantative approach used by Gabor to analyse sound signals differs from an elementary particle in quantum physics. He only wanted to determine an elementary particle in terms of the human perception of sound. He was interested in the threshold of the human ear, and what one perceives as a single elementary sound signal. After determining 21ms at 500Hz to be about the shortest duration of sound perceivable he set about looking at ways to capture and reproduce sounds of that duration. Applying a quantative methodology to sound analysis put him at the disposal of all quantum research that had been conducted over the past 40 years. Through a mathematical process involving FT, time-domain and quantum analysis, he quantised the elementary sound and created the analogy of the sound atom [10].

Gabor's next step was to use this information in a practical sense. He created some machines that could granulate and reconstitute a sound, although only in the time domain, which is where granular synthesis has mostly resided. The most popular of these machines was a kinematic frequency converter based on a tape device, but with multiple rotating heads that could be used to change the time domain of a sound by stretching or condensing the time base without affecting the pitch, or for changing the pitch without affecting the time base [10].

In 1950 Werner Meyer-Eppler gave a talk entitled *Das Klangfarbenproblem der elektronischen Musik* (*The Problem of Timbre in Electronic Music*) in which he made use of Gabor's ideas on musical quanta [26]. Stockhausen, Pierre Schaeffer, and Jacques Poullin were all in contact with Meyer-Eppler during this period, which obviously prompted them to start thinking about this concept. Stockhausen began working in his poorly equipped studio in Paris trying to superimpose sine waves on top of each other, and cut them down to size without even having access to a tape recorder [17]. Schaeffer and Poullin set about creating a similar device to Gabor. They named their device the *Phonogéne*. Later in the 1960s, Herbert Eimert used a similar device created by the company Springer called the *Tempophon*. Whilst their work in that time is interesting, it was Xenakis who came up with the next conceptual idea (See Figure 2).

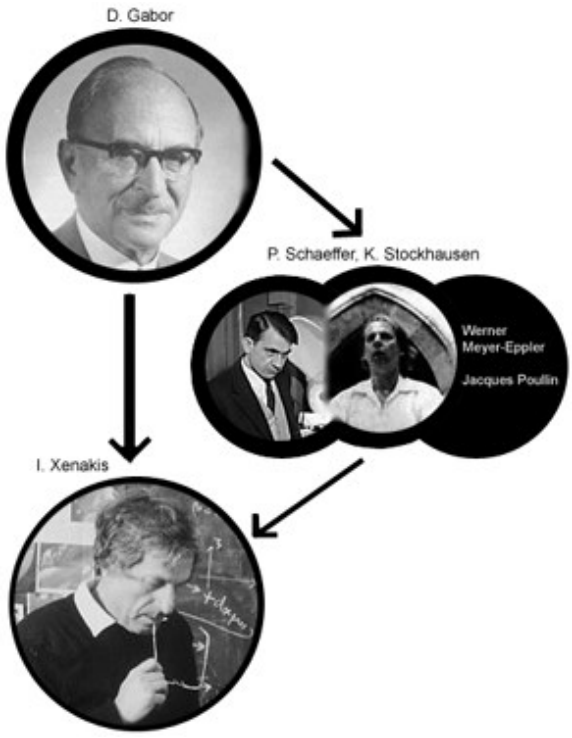


Figure 2: Scientific knowledge becoming a compositional tool.

3 Composition: Splicing

It is Iannis Xenakis who first conceived and executed a composition using granular synthesis. He went through Gabor's paper and then set about formulating an idea which takes the Gabor Matrix and turns it into a cube, a grid of Gabor matrices, which he commonly referred to as a screen. The cube was used for setting up 3D spatialised clouds of sound [33].

In 1959 he used this process for creating the piece entitled *Analogique B*. The clouds were determined stochastically using Markovian indeterminism, following a detailed method outlined in his book *Formalized Music*. Xenakis designated the grains using a number of different array matrices of parameters on a screen, the screen representing the audio and time spectrum. Each cell of the screen was given parameters associated to the grain structure in that audio-time region. Xenakis created books of screens to represent complex audio sounds. The sound source he used for this composition was recorded sine tones generated by tone generators onto analogue tape. The tape was spliced into 40ms fragments and then meticulously pieced together. *Analogique B* was created to complement *Analogique*

A, which was derived using the same stochastic methods, but written for two violins, two violoncellos, and two contrabasses [34][26]. Interestingly, Schaeffer was openly opposed to Analogique B because he didn't like the sound of it, and he did not like the fact that the composition was so scientifically based [19].

Xenakis' book *Formalized Music* and his workshops moved granular synthesis to the next stage; from a knowledge base, through a compositional idea, leading to a computer process.

4 Computer Process: Klang!

Xenakis' efforts to compose with granular synthesis by tape splicing are to be highly commended, but in order to make granular synthesis a more widespread tool for composition it needed to have a digital medium with which to manipulate the grains. The computer made it possible to incorporate many mathematical algorithms directly into the structuring process of the granular texture. In the early days of computing however this was still a very laborious task.

Motivated by Xenakis, the composer Curtis Roads became extremely interested in granular synthesis [22]. Roads first heard about granular synthesis when he attended a workshop conducted by Iannis Xenakis in 1972. Since first learning about granular synthesis, Roads has spent much time researching and writing on the topic [23][24][25][26] (See Figure 3).

Roads first began working with computer sound synthesis on an advanced mainframe, the dual-processor Burroughs B6700. This mainframe was exceptionally good for its time, but using it was still very time consuming. The Burroughs machine was running the program Music V, written in 1966 by Max Mathews, and had to be programmed using the language Algol [21]. All input was done through the use of punched paper cards. Due to storage limits on the mainframe, Roads was limited to one minute of monaural sound at a sampling rate of 20 kHz. It would take several days to process a minute of sound. The mainframe, as was typical, was also unable to produce audio. Roads would take the punch cards to the Burroughs machine. Once it had finished processing, the data would be written to a (large) digital tape. The digital tape would then have to be transferred to a disk cartridge which involved setting up an appointment at the Scripps Institute of Oceanography. Finally, the cartridge could be converted to Audio on a DEC PDP-11/20 at the Centre for Music Experiment [18][26].

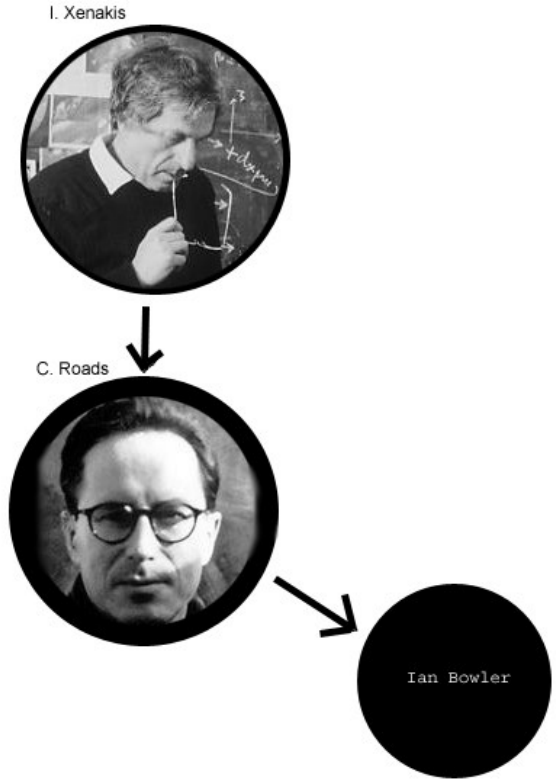


Figure 3: Composing on computers

Using this process Curtis Roads composed the first piece of computer generated music using granular synthesis. It was just a technical experiment entitled Klang-1, written in 1974. This piece of music had a duration of 30 seconds. It was an experiment testing three parameters: Grain Envelope, grain duration and density. Klang-1 featured 766 grains of sound. Each grain needed to be programmed onto a separate punch card. Roads treated each grain as if it were an individual note, just with a very short duration; specifying the frequency, start time and duration of each note. Roads fixed the duration of each grain at 40ms, which was identical to the grain duration of Xenakis' Analogique B. The densities ranged from 0 to 25 grains per second. The grain envelope was a simple Gaussian envelope. The grains were all generated as sine waves with frequencies from a 1000-tone scale ranging from 16.11Hz to 9937.84Hz. At the completion of this experiment of granular synthesis on a computer Roads made the following statement:

> I did not intend *Klang-1* to be anything more than a technical experiment. Yet I vividly recall the magical impression that the granular sound had on me as it poured forth from the computer for the first time [26].

The second piece of music, using granular synthesis, realised on a computer, was entitled *Prototype* and was written in 1975. This was an actual composition rather than an experiment. To compose this piece Roads wrote a program called *PLFKLANG* in *Algol* which could generate thousands of grain specifications based on seven high-level cloud parameters. Defining parameters as clouds is significant because it can define an entire texture of grains, whereas the grain is discrete as an individual particle [26].

Later versions of the computer program *Music V,* maintained and developed by Daniel Arfib, contained probably the first code written to perform analysis-synthesis in terms of the Gabor grain. Risset used the granular analysis-synthesis tools of Music V, on an IBM-PC, to perform time stretching of recorded samples in his piece entitled *Attracteurs Etranges* [22].

Following on from the work of Roads, Ian Bowler in 1985 proposed a process called Switched Waveform Synthesis, based on granular synthesis. Its purpose was to create different timbres within the constraints of a microcomputer [3]. This idea however, did not get developed very far.

5 Real-Time: Anytime

The next conceptual idea for granular synthesis came in 1986 through Barry Truax. Truax wrote a computer program that allowed him to use granular synthesis in real-time. It was a breakthrough that has made granular synthesis a much easier process to work with. It led the way to a barrage of programs that all focus on automating various parts of the granular synthesis process, allowing it to be played back in real-time (See figure 4).

Barry Truax started learning about granular synthesis after reading an article published in the *Computer Music Journal* by Roads in 1978 entitled *Automated Granular Synthesis of Sound* [23]. Truax's approach to granular synthesis was very different to that of Roads. Truax pioneered real-time granular synthesis right from the onset of his research. In 1986 he wrote a real-time application of granular synthesis implemented using his previously developed *POD & PODX* system. At the time Truax implemented the real-time granular synthesis algorithms in *PODX* he was using a DMX-1000 Digital Signal Processor controlled by a DEC LSI-11/23 microcomputer. Using both the DMX-1000 and the DEC LSI-11/23 meant that the sound generation could be handled by a processor optimized for audio, whilst the microcomputer would only have to be concerned with number-crunching and outputting a stream of numbers [27][28][30][31]. The computer was set up so that the granular synthesis parameters could be changed in real-time with a single key stroke. Truax was also able to store a range of presets and assign them to any unused key on the keyboard. The parameters could also be set using a *tendency mask*, which is a graphical representation of a range over time. The graphical tendencies masks would be processed as a combination of presets and ramps [29]. *Riverrun* was the first piece of granular synthesis based music Truax composed using the PODX real-time system.

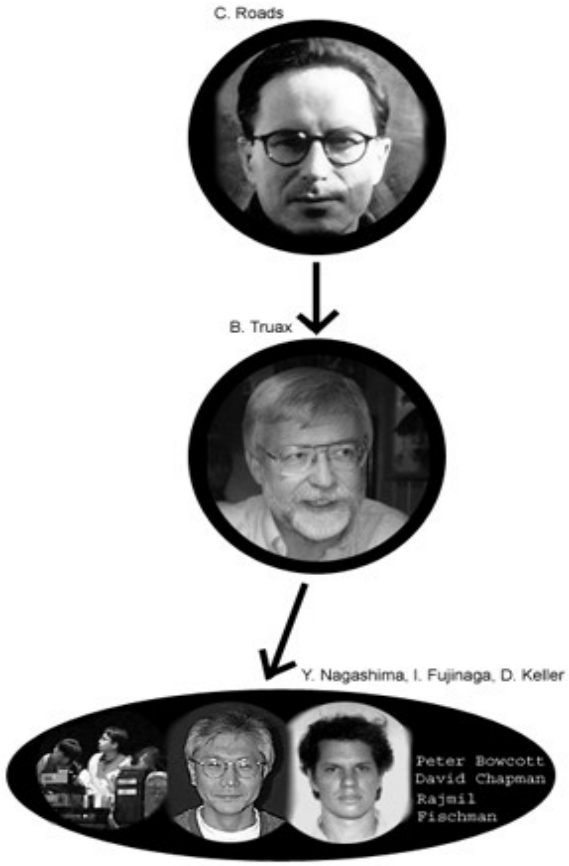


Figure 4: Computer Extensions

A most notable feature of the work by Barry Truax was his attention to control. The PODX system was the first in a whole wave of different control ideas for automated granular synthesis on a computer. There have been many other control methods designed over the last 15 years by a large range of composers.

Using genetic algorithms as a control method for grain distribution has been experimented with by Peter Bowcott, Ichiro Fujinaga, Horner, M. Hamman, and Goldberg [12]. Bowcott based his genetic model on Cellular Automation [2]. Bowcott has also used population and growth models upon which to control grain distribution [1]. David Chapman, Michael Clarke, Martin Smith, and Paul Archbold researched and created a computer language called *AMY* to help

them control and study fractal based granular synthesis [4]. Ecologically-based granular synthesis, as a means of control, has been experimented with by Damián Keller and Barry Truax. What is interesting in their approach to sound organisation is their view of what they are trying to achieve. They pointed to a statement made by Dannenberg to the effect that there is a lack of research done in sound organisation for time scales ranging from 100ms to a several seconds. The micro-scale synthesis issues have been well researched, such as synthesis and analysis of single acoustical quanta. What Keller and Truax began researching were short recognisable natural sounds, such as bouncing, scraping, breaking and filling. They created algorithms to create these types of sounds in *Csound* and *Cmask* using granular synthesis [16]. Yoichi Nagashima has used neural networks to automate and refine the control of grain distribution in a project called PEGASUS [20]. Rajmil Fischman has been experimenting with the Schrödinger equation as a control method for grain distribution [8].

As we can see, there is an abundance of ideas for control and creation of granular synthesis available. The conception and continuity of granular synthesis has been uncovered. The question then becomes, how do we ourselves make use of these ideas? Where are these ghosts actually revealed?

If we are computer programmers, we can simply begin assimilating the maths and science and start programming. There are even some programming languages and libraries available, which will allow you to create your own program, but still work within a framework that simplifies a lot of common commands. A program such as *jMusic* gives very low level control of the sound and implementation structure. Some higher level programs and music composition languages available include *Supercollider, Csound, Max/MSP, AudioMulch*, and *Reaktor*.

For general usage and performance, setting up an entire programming language to perform a simple task might be like using a sledge hammer to crack a nut. In these cases there are a plethora of applications that perform specific granular synthesis tasks.

An extremely powerful program for time stretching using granular synthesis is *MacPOD*, based on the *GSAMX* functionality of *PODX* [30], and written by Damián Keller and Chris Rolfe for Macintosh [14][15]. The algorithmic compositional program called *AL & ERWIN* (AL for Algorithm and Erwin as in Erwin Schrödinger) is a great tool for experimenting with granular clouds based on the Schrödinger equation from quantum mechanics [7]. *Amber* by Matthew Mccabe is a program designed specifically for creating interesting and complex sounds [35]. *Cloud Generator* by Curtis Roads and John Alexander is a non real-time program for creating simple granular clouds and makes a good learning tool [36]. *Granulator* by Nicholas Fournel, and *Granulab* by Rasmus Ekman both allow the composer to experiment with granular synthesis in real-time. They give the composer control over the standard parameters such as the shape and size of the grains, and also the density of the overall texture [37][38].

With this array of programs available, and many more not even mentioned, it is no wonder that granular synthesis is seeping into the commercial music world. DJ Spooky and Fatboy Slim have both made good use of granular synthesis, especially with Fatboy Slim's, *The Rockafeller Skank,* making the pop charts across the world [6].

There are still more branches to add to this tree of intellectual and compositional knowledge in granular synthesis. They may be small branches, such as control algorithms, and implementation programs, or they may be larger conceptual ideas. Are we happy just using the branches currently available, or do we want to create our own branches?

6 Conclusion

Through the combination of the Fourier Transform and quantum physics, Dennis Gabor has allowed the idea of the sound atom to become a real building block in modern music. It was initially just a concept for communication, but through the work of Xenakis, Roads, Truax, and many others, it has become a compositional and musical tool with a large spectrum of uses.

As the concept of granular synthesis is developed further, there will be more ghosts, in the granular synthesis machine, that will help drive future programs and ideas.

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